stats

Linda Herrera

Packages

```
library(tidyr)
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
library(ggsignif)
library(gtsummary)
library(dplyr)
library(rstatix)
library(ggpp)
library(lmerTest)
library(broom)
library(broom.mixed)
library(purrr)
library(gt)
```

Data

```
# Data Frame with ALL sites, TC (utc and tc isco's), CC, IT
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
```

```
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date)) # telling r the date format

# Trail Creek site order
site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

# Trail Creek data
trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

# Coal Creek data
coal_creek <- compiled_isotope |>
 filter(location %in% c("cc"))

# Italian Creek data
italian_creek <- compiled_isotope |>
  filter(location == "it")

# Trail Creek ISCO data combined
trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date))# telling r the date format

# Upstream Trail Creek ISCO
utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

# Downstream Trail Creek ISCO
tc_isco <- trail_creek_isco |>
   filter(site == "tc_isco")

# Trail Creek gradient data
```

```
gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

# Coal Creek gradient data
gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

# Italian Creek gradient data
gradient_it <- read_csv(here("data", "it_gradient_data.csv"))

# ALL Discharge data
discharge_isco <- read_csv(here("data", "isco_discharge_lc.csv")) |>
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    period = if_else(date < as.Date("2026-05-14"), "pre-peak", "post-peak") # before & after
  )

# Upstream Trail Creek ISCO
discharge_utc_isco <- discharge_isco |>
  filter(isco %in% c("utc"))

# Downstream Trail Creek ISCO
discharge_tc_isco <- discharge_isco |>
  filter(isco %in% c("tc"))

# Upstream and Downstream Trail Creek ISCO
us_ds <- read_csv(here("data", "us_ds.csv")) |>
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    creek = case_when(
      grepl("^cc", location) ~ "Coal Creek",
      grepl("^it", location) ~ "Italian Creek",
      location %in% c("utc", "tc") ~ "Trail Creek"),
    reach = case_when(
      location %in% c("cc-us", "it-us", "utc") ~ "upstream",
      location %in% c("cc-ds", "it-ds", "tc") ~ "downstream"))

# Summary stats per period (per reach, if you have upstream/downstream)
discharge_stat <- discharge_isco |>
  group_by(period) |> # add reach here: group_by(reach, period)
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE),
```

```
    median_lc = median(lc_excess, na.rm = TRUE)
  )
```

Statistical Tests

```
# ANOVA,
# lc-excess and location
anova <- aov(lc_excess ~ location, data = compiled_isotope)
summary(anova)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)    
location      3  209.2   69.72   36.53 <2e-16 ***
Residuals   111  211.8    1.91                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# TukeyHSD, compares every possible pair of group means
TukeyHSD(anova)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = lc_excess ~ location, data = compiled_isotope)

$location
             diff        lwr        upr     p adj
it-cc   2.2952900  0.8997570  3.6908230 0.0002219
tc-cc  -1.8984083 -3.0613525 -0.7354642 0.0002508
utc-cc -0.7528817 -1.9388567  0.4330933 0.3520168
tc-it  -4.1936983 -5.2595543 -3.1278424 0.0000000
utc-it -3.0481717 -4.1391100 -1.9572333 0.0000000
utc-tc  1.1455267  0.3741168  1.9169366 0.0010318
```

```
# Trail Creek
# lc-excess & normalized stream distance

# 6/24 - lc-excess & normalized stream distance
model.1 <- lm(`lc-excess_6-24` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.1)
```

```
Call:
lm(formula = `lc-excess_6-24` ~ normalized_stream_distance_m,
    data = gradient_tc)

Residuals:
       1        2        3        4        5        6        7 
-0.36861 -0.32358  0.58444  0.03641  0.15093  0.31789 -0.39748 

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)    
(Intercept)                  -1.3907116  0.2562773  -5.427 0.002880 ** 
normalized_stream_distance_m -0.0011403  0.0001537  -7.417 0.000702 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.416 on 5 degrees of freedom
Multiple R-squared:  0.9167,    Adjusted R-squared:    0.9 
F-statistic: 55.01 on 1 and 5 DF,  p-value: 0.0007016
```

```r
# 6/30 - lc-excess & normalized stream distance
model.2 <- lm(`lc-excess_6-30` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.2)
```

```
Call:
lm(formula = `lc-excess_6-30` ~ normalized_stream_distance_m,
    data = gradient_tc)

Residuals:
       1        2        3        4        5        6        7 
-0.08528 -0.20669  0.24371  0.25137  0.19568 -0.61753  0.21874 

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)    
(Intercept)                  -2.2800413  0.2204137  -10.34 0.000145 ***
normalized_stream_distance_m -0.0008555  0.0001322   -6.47 0.001314 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3578 on 5 degrees of freedom
Multiple R-squared:  0.8933,    Adjusted R-squared:  0.872 
```

```
F-statistic: 41.86 on 1 and 5 DF,  p-value: 0.001314
```

```r
# 7/15 - lc-excess & normalized stream distance
model.3 <- lm(`lc-excess_7-15` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.3)
```

```

Call:
lm(formula = `lc-excess_7-15` ~ normalized_stream_distance_m,
    data = gradient_tc)

Residuals:
       1        2        3        4        5        6        7
-0.36655 -0.34573  0.61325  0.47415 -0.07123 -0.28328 -0.02060

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                  -1.1034551  0.2671852  -4.130  0.00909 **
normalized_stream_distance_m -0.0010277  0.0001603  -6.412  0.00137 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4337 on 5 degrees of freedom
Multiple R-squared:  0.8916,    Adjusted R-squared:  0.8699
F-statistic: 41.11 on 1 and 5 DF,  p-value: 0.001369
```

```r
# Coal Creek
# lc-excess & normalized stream distance

# 6/23 - lc-excess & normalized stream distance
model.4 <- lm(`lc-excess_6-23` ~ normalized_stream_distance_m, data = gradient_cc)
summary(model.4)
```

```

Call:
lm(formula = `lc-excess_6-23` ~ normalized_stream_distance_m,
    data = gradient_cc)

Residuals:
      1       2       3       4       5
-0.3323  0.6279  0.2566 -0.3412 -0.2109
```

```
Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                  1.5393266  0.4492095   3.427   0.0416 *
normalized_stream_distance_m -0.0006445  0.0002206  -2.921   0.0614 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4938 on 3 degrees of freedom
Multiple R-squared:  0.7399,    Adjusted R-squared:  0.6532
F-statistic: 8.533 on 1 and 3 DF,  p-value: 0.06144
```

```r
# 7/7 - lc-excess & normalized stream distance
model.5 <- lm(`lc-excess_7-7` ~ normalized_stream_distance_m, data = gradient_cc)
summary(model.5)
```

```
Call:
lm(formula = `lc-excess_7-7` ~ normalized_stream_distance_m,
    data = gradient_cc)

Residuals:
      1       2       3       4       5
 0.2092  0.3569 -0.8765 -0.8080  1.1184

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)
(Intercept)                 -0.333192   0.885667  -0.376   0.7318
normalized_stream_distance_m -0.001031   0.000435  -2.369   0.0986 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9735 on 3 degrees of freedom
Multiple R-squared:  0.6517,    Adjusted R-squared:  0.5356
F-statistic: 5.613 on 1 and 3 DF,  p-value: 0.09859
```

```r
# Italian Creek
# lc-excess & normalized stream distance

# 7/1 - lc-excess & normalized stream distance
```

```r
model.6 <- lm(`lc_excess_7_1` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.6)
```

```
Call:
lm(formula = lc_excess_7_1 ~ normalized_stream_distance_m, data = gradient_it)

Residuals:
        1         2         3         4         5 
 0.190779 -0.256938 -0.069233  0.141983 -0.006591 

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)    
(Intercept)                  3.549e+00  1.634e-01  21.715 0.000214 ***
normalized_stream_distance_m -9.279e-05  8.643e-05  -1.074 0.361692    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2061 on 3 degrees of freedom
Multiple R-squared:  0.2775,    Adjusted R-squared:  0.03673 
F-statistic: 1.153 on 1 and 3 DF,  p-value: 0.3617
```

```r
# 7/16 - lc-excess & normalized stream distance

model.7 <- lm(`lc-excess_7_16` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.7)
```

```
Call:
lm(formula = `lc-excess_7_16` ~ normalized_stream_distance_m, 
    data = gradient_it)

Residuals:
       1        2        3        4        5 
 0.19695 -0.22076 -0.11740  0.08273  0.05848 

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)   
(Intercept)                  1.438e+00  1.529e-01   9.403  0.00255 **
normalized_stream_distance_m -3.170e-04  8.088e-05  -3.919  0.02954 * 
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1928 on 3 degrees of freedom
Multiple R-squared:  0.8366,    Adjusted R-squared:  0.7822
F-statistic: 15.36 on 1 and 3 DF,  p-value: 0.02954
```

```
# 7/23 - lc-excess & normalized stream distance

model.8 <- lm(`lc_excess_7_23` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.8)
```

```
Call:
lm(formula = lc_excess_7_23 ~ normalized_stream_distance_m, data = gradient_it)

Residuals:
       1        2        3        4        5 
 0.18103 -0.20000 -0.05687 -0.04930  0.12514 

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)   
(Intercept)                   1.005e+00  1.405e-01   7.154  0.00563 **
normalized_stream_distance_m -2.609e-04  7.428e-05  -3.512  0.03914 * 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1771 on 3 degrees of freedom
Multiple R-squared:  0.8044,    Adjusted R-squared:  0.7391
F-statistic: 12.33 on 1 and 3 DF,  p-value: 0.03914
```

```
# Trail Creek
# lc-excess & cumulative dams

gradient_tc$cumulative_dams <- as.numeric(as.character(gradient_tc$cumulative_dams))

# 6/24 - lc-excess & cumulative dams
model.9 <- lm(`lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
summary(model.9)
```

```
Call:
```

```
lm(formula = `lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)

Residuals:
      2       3       4       5       6       7 
-0.4806  0.1654  0.1151  0.4332  0.4826 -0.7157 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)   
(Intercept)     -1.613391   0.362589  -4.450  0.01125 * 
cumulative_dams -0.034669   0.006788  -5.107  0.00695 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5487 on 4 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.867, Adjusted R-squared:  0.8338 
F-statistic: 26.08 on 1 and 4 DF,  p-value: 0.006948
```

```r
# 6/30 - lc-excess & cumulative dams
model.10 <-  lm(`lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
summary(model.10)
```

```
Call:
lm(formula = `lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)

Residuals:
       2        3        4        5        6        7 
-0.27039 -0.01815  0.34655  0.43245 -0.47798 -0.01249 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     -2.501209   0.257870  -9.699 0.000632 ***
cumulative_dams -0.025481   0.004828  -5.278 0.006179 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3902 on 4 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.8744,	Adjusted R-squared:  0.843 
F-statistic: 27.86 on 1 and 4 DF,  p-value: 0.006179
```

```r
# 7/15 - lc-excess & cumulative dams
model.11 <- lm(`lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)
summary(model.11)
```

```
Call:
lm(formula = `lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)

Residuals:
       2        3        4        5        6        7 
-0.55012  0.17680  0.52850  0.19513 -0.09976 -0.25056 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)   
(Intercept)     -1.241283   0.281127  -4.415  0.01155 * 
cumulative_dams -0.032606   0.005263  -6.195  0.00345 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4254 on 4 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.9056,    Adjusted R-squared:  0.882 
F-statistic: 38.38 on 1 and 4 DF,  p-value: 0.003452
```

```r
# Coal Creek
# lc-excess & cumulative dams

gradient_cc$cumulative_dams <- as.numeric(as.character(gradient_cc$cumulative_dams))

# 6/23 - lc-excess & cumulative dams
model.12 <-  lm(`lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)
summary(model.12)
```

```
Call:
lm(formula = `lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)

Residuals:
       2        3        4        5 
 0.11851  0.07466 -0.40553  0.21236 
```

```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.69377    0.41827   4.049   0.0559 .
cumulative_dams -0.04687    0.01195  -3.921   0.0593 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3385 on 2 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.8849,    Adjusted R-squared:  0.8273
F-statistic: 15.37 on 1 and 2 DF,  p-value: 0.05931
```

```
# 7/7 - lc-excess & cumulative dams
model.13 <-  lm(`lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)
summary(model.13)
```

```
Call:
lm(formula = `lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)

Residuals:
      2       3       4       5
 0.8361 -0.6905 -0.9591  0.8135

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -2.13831    1.45089  -1.474    0.278
cumulative_dams -0.01660    0.04146  -0.400    0.728

Residual standard error: 1.174 on 2 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.0742,    Adjusted R-squared:  -0.3887
F-statistic: 0.1603 on 1 and 2 DF,  p-value: 0.7276
```

```
# Trail Creek
# lc-excess & cumulative dam length
gradient_tc$cumulative_dam_length <- as.numeric(as.character(gradient_tc$cumulative_dam_leng

# 6/24 - lc-excess & cumulative dam length
model.14 <- lm(`lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)
summary(model.14)
```

```
Call:
lm(formula = `lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)

Residuals:
      2       3       4       5       6       7 
-0.3782  0.3121 -0.2486  0.3477  0.3552 -0.3882 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)   
(Intercept)           -1.715833   0.259857  -6.603  0.00273 **
cumulative_dam_length -0.056153   0.008125  -6.911  0.00230 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4183 on 4 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.9227,    Adjusted R-squared:  0.9034 
F-statistic: 47.77 on 1 and 4 DF,  p-value: 0.002299
```

```r
# 6/30 - lc-excess & cumulative dam length
model.15 <-  lm(`lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)
summary(model.15)
```

```
Call:
lm(formula = `lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)

Residuals:
       2        3        4        5        6        7 
-0.16963  0.11233  0.09129  0.36273 -0.58878  0.19207 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)           -2.601968   0.233442  -11.15 0.000369 ***
cumulative_dam_length -0.040214   0.007299   -5.51 0.005295 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3758 on 4 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.8836,    Adjusted R-squared:  0.8545 
```

```
F-statistic: 30.36 on 1 and 4 DF,  p-value: 0.005295
```

```r
# 7/15 - lc-excess & cumulative dam length
model.16 <- lm(`lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)
summary(model.16)
```

```
Call:
lm(formula = `lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)

Residuals:
      2       3       4       5       6       7 
-0.4429  0.3245  0.1916  0.1118 -0.2269  0.0420 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)   
(Intercept)           -1.348532   0.197391  -6.832  0.00240 **
cumulative_dam_length -0.052360   0.006172  -8.484  0.00106 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3178 on 4 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.9474,    Adjusted R-squared:  0.9342 
F-statistic: 71.97 on 1 and 4 DF,  p-value: 0.001058
```

```r
# Coal Creek
# lc-excess & cumulative dam length

gradient_cc$cumulative_dam_length <- as.numeric(as.character(gradient_cc$cumulative_dam_leng

# 6/23 - lc-excess & cumulative dam length
model.17 <-  lm(`lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)
summary(model.17)
```

```
Call:
lm(formula = `lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)

Residuals:
       2        3        4        5 
```

```
 0.10405 -0.06714 -0.20308  0.16617

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)            1.547568   0.225772   6.855   0.0206 *
cumulative_dam_length -0.027556   0.004094  -6.730   0.0214 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2052 on 2 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.9577,    Adjusted R-squared:  0.9366
F-statistic: 45.29 on 1 and 2 DF,  p-value: 0.02137
```

```r
# 7/7 - lc-excess & cumulative dam length
model.18 <-  lm(`lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)
summary(model.18)
```

```
Call:
lm(formula = `lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)

Residuals:
      2       3       4       5
 0.7792 -0.7622 -0.8651  0.8481

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           -2.11270    1.26808  -1.666    0.238
cumulative_dam_length -0.01134    0.02300  -0.493    0.671

Residual standard error: 1.152 on 2 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.1083,    Adjusted R-squared:  -0.3375
F-statistic: 0.243 on 1 and 2 DF,  p-value: 0.6709
```

```r
# Trail Creek ISCO's
# Correlation test of discharge and lc-excess

# Downstream Trail Creek ISCO
cor.test(discharge_tc_isco$discharge, discharge_tc_isco$lc_excess)
```

```
	Pearson's product-moment correlation

data:  discharge_tc_isco$discharge and discharge_tc_isco$lc_excess
t = 24.555, df = 44, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9379629 0.9808167
sample estimates:
      cor 
0.9653947 
```

```r
# Upstream Trail Creek ISCO
cor.test(discharge_utc_isco$discharge, discharge_utc_isco$lc_excess)
```

```
	Pearson's product-moment correlation

data:  discharge_utc_isco$discharge and discharge_utc_isco$lc_excess
t = 4.7379, df = 38, p-value = 3e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3676755 0.7739744
sample estimates:
      cor 
0.6093889 
```

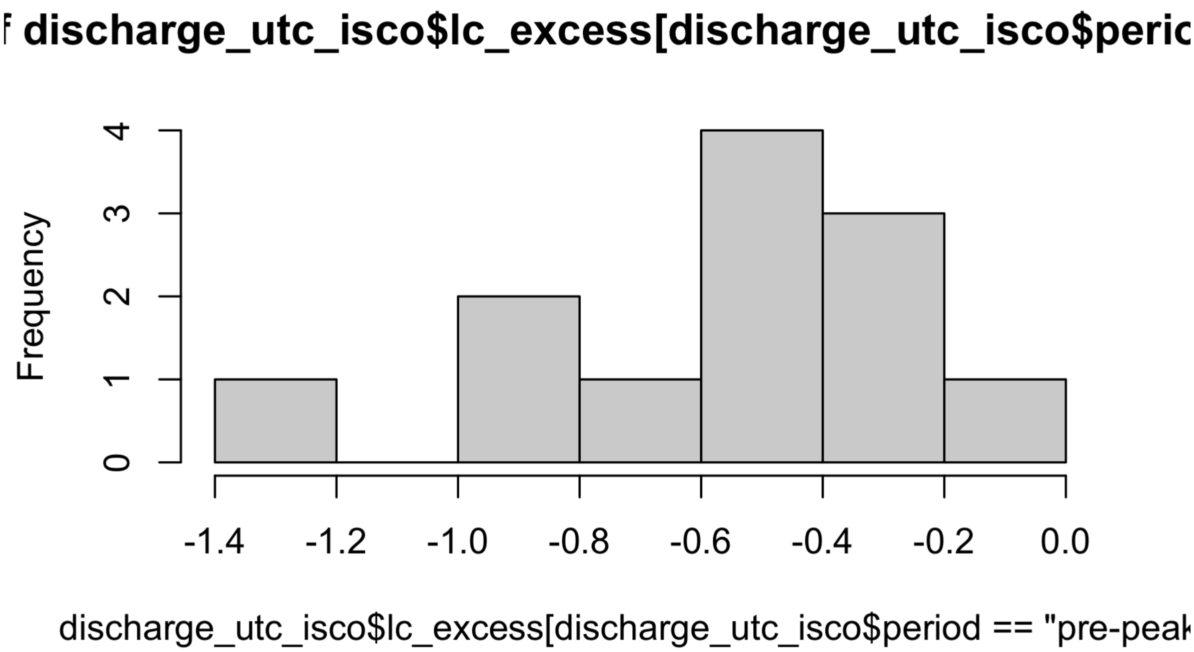
```r
# Downstream Trail Creek - lc-excess & discharge
lm(lc_excess ~ discharge, data = discharge_tc_isco)
```

```
Call:
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)

Coefficients:
(Intercept)    discharge  
     -5.986       42.209  
```

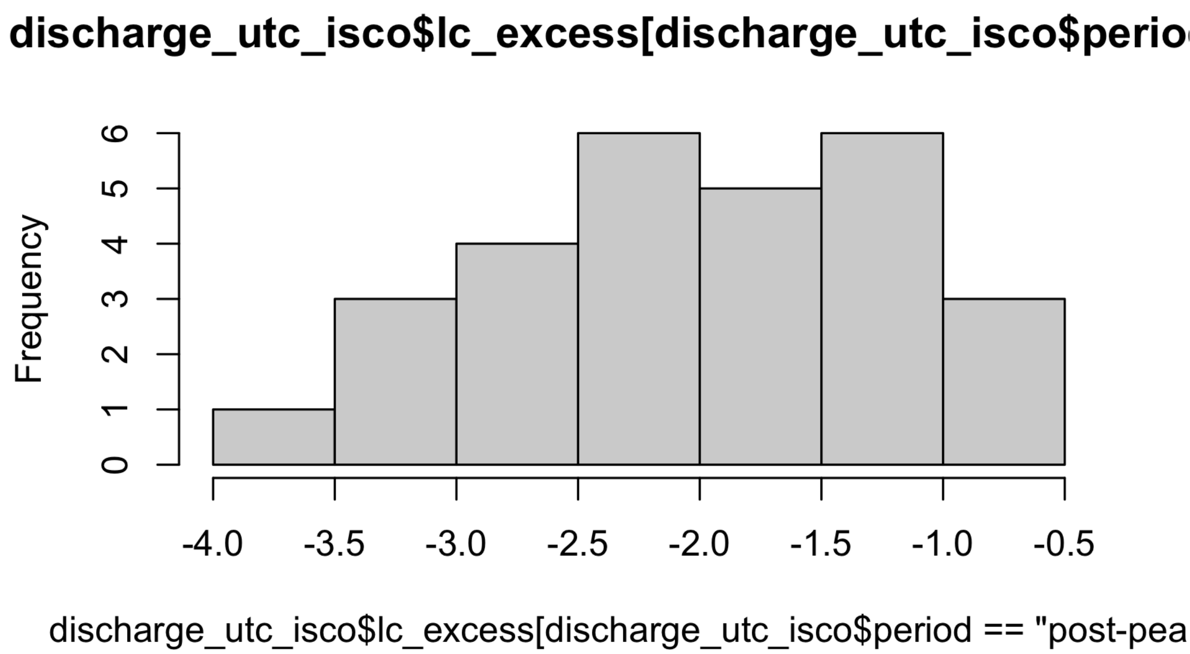
```r
# Upstream Trail Creek - lc-excess & discharge
lm(lc_excess ~ discharge, data = discharge_utc_isco)
```

```
Call:
lm(formula = lc_excess ~ discharge, data = discharge_utc_isco)

Coefficients:
(Intercept)    discharge
     -2.975       19.597
```

```
# Visual check
hist(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"]) # skewed left
```




```
hist(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"]) # normal distril
```




```
# Formal test (Shapiro-Wilk) - only reliable with reasonable sample size
shapiro.test(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"])
```

```
    Shapiro-Wilk normality test

data:  discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"]
W = 0.97691, p-value = 0.9683
```

```
shapiro.test(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"])
```

```
    Shapiro-Wilk normality test

data:  discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"]
W = 0.96427, p-value = 0.4377
```

```r
# Parametric alternative (Welch's t-test, doesn't assume equal variance) - use if normal
t.test(lc_excess ~ period, data = discharge_utc_isco)
```

```

	Welch Two Sample t-test

data:  lc_excess by period
t = -8.1949, df = 37.963, p-value = 6.38e-10
alternative hypothesis: true difference in means between group post-peak and group pre-
peak is not equal to 0
95 percent confidence interval:
 -1.784032 -1.077199
sample estimates:
mean in group post-peak  mean in group pre-peak
             -1.9979821              -0.5673667
```

```r
discharge_utc_isco |>
  group_by(period) |>
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    median_lc = median(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE)
  )
```

```
# A tibble: 2 x 5
  period        n mean_lc median_lc sd_lc
  <chr>     <int>   <dbl>     <dbl> <dbl>
1 post-peak    28  -2.00     -2.02  0.771
2 pre-peak     12  -0.567    -0.542 0.333
```

```r
var.test(lc_excess ~ period, data = discharge_utc_isco)
```

```

	F test to compare two variances

data:  lc_excess by period
F = 5.3505, num df = 27, denom df = 11, p-value = 0.005712
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
```

```
  1.702796 13.452718
sample estimates:
ratio of variances
          5.350494
```

```r
# Correlation of the lc-excess & location

compiled_isotope$date_c <- as.numeric(compiled_isotope$date) - mean(as.numeric(compiled_isot
# center the date variable by subtracting its mean.

model_location <- lmer(lc_excess ~ location + date_c + (1 | site), data = compiled_isotope)
# fixed effects: location differences + linear time trend
# random intercept: allows each site to have its own baseline lc-excess, accounting for
# the ~36% of variance that's due to site-level clustering rather than location or date
summary(model_location)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: lc_excess ~ location + date_c + (1 | site)
   Data: compiled_isotope

REML criterion at convergence: 334.7

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.18174 -0.59314 -0.05398  0.65138  2.16327

Random effects:
 Groups   Name        Variance Std.Dev.
 site     (Intercept) 0.4233   0.6506
 Residual             0.8648   0.9300
Number of obs: 115, groups:  site, 20

Fixed effects:
             Estimate Std. Error        df t value Pr(>|t|)
(Intercept)   0.28574    0.38862  25.19091   0.735   0.4690
locationit    2.91690    0.53729  19.55019   5.429 2.79e-05 ***
locationtc   -2.88718    0.52093  16.75436  -5.542 3.77e-05 ***
locationutc  -1.55173    0.54866  15.90779  -2.828   0.0122 *
date_c       -0.04662    0.00458 109.92129 -10.179  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
             (Intr) loctnt locationtc locationutc
locationit   -0.656
locationtc   -0.722  0.501
locationutc  -0.692  0.473  0.506
date_c       -0.236 -0.114  0.073      0.099
```

```
# centering doesn't affect slopes, only the intercept's ^ from the (1|site)

anova(model_location)
```

```
Type III Analysis of Variance Table with Satterthwaite's method
          Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
location 111.828  37.276     3  15.082  43.102 1.238e-07 ***
date_c    89.602  89.602     1 109.921 103.605 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# the enrichment effect consistent across geologically distinct sites? This is a reach × cree

model_reach <- lmer(lc_excess ~ reach * creek + (1 | date), data = us_ds)
summary(model_reach)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: lc_excess ~ reach * creek + (1 | date)
   Data: us_ds

REML criterion at convergence: 27.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.05323 -0.31166 -0.03435  0.35767  0.96783

Random effects:
 Groups   Name        Variance Std.Dev.
 date     (Intercept) 1.19749  1.0943
 Residual             0.09793  0.3129
Number of obs: 16, groups:  date, 8
```

```
Fixed effects:
                                 Estimate Std. Error      df t value Pr(>|t|)
(Intercept)                       -1.3596     0.8048  5.3922  -1.689  0.14768
reachupstream                      1.9010     0.3129  5.0000   6.075  0.00175
creekItalian Creek                 2.7533     1.0390  5.3922   2.650  0.04217
creekTrail Creek                  -3.3642     1.0390  5.3922  -3.238  0.02064
reachupstream:creekItalian Creek  -1.1078     0.4040  5.0000  -2.742  0.04069
reachupstream:creekTrail Creek     1.0219     0.4040  5.0000   2.530  0.05256

(Intercept)
reachupstream                    **
creekItalian Creek               *
creekTrail Creek                 *
reachupstream:creekItalian Creek *
reachupstream:creekTrail Creek   .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) rchpst crkItC crkTrC rch:IC
reachupstrm -0.194
crekItlnCrk -0.775  0.151
creekTrlCrk -0.775  0.151  0.600
rchpstrm:IC  0.151 -0.775 -0.194 -0.117
rchpstrm:TC  0.151 -0.775 -0.117 -0.194  0.600
```

```r
anova(model_reach)
```

```
Type III Analysis of Variance Table with Satterthwaite's method
             Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
reach       13.5224 13.5224     1     5 138.082 7.849e-05 ***
creek        3.0212  1.5106     2     5  15.425  0.007264 **
reach:creek  3.4032  1.7016     2     5  17.376  0.005611 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# the lmer() model is more powerful here because it pools information across all three creeks


# What the model is actually telling you:
# `reach` main effect (p = 7.8e-05): across all sites, upstream vs. downstream location matte
# `creek` main effect (p = 0.0073): the three creeks differ in baseline lc-excess overall, wh
```

```
# `reach:creek` interaction (p = 0.0056): this is the key term for aim (ii). It's significan
# Looking at the coefficients:
# Italian Creek's interaction term is −1.11 (p = 0.041) - the upstream/downstream gap is smal

# Italian Creek is your undammed control, so a weaker reach effect there strengthens your cas

# Trail Creek's interaction term is +1.02 (p = 0.053, right at the edge) - trending toward a
```

```
# ==============================================================
# Compile all statistical tests into publication-ready tables
# ==============================================================
# NOTE: run this AFTER your existing stats.R script, so all model
# objects (anova, model.1-18, discharge_*_isco tests, us_ds model, etc.)
# already exist in your environment.

# --------------------------------------------------------------
# TABLE 1. Pairwise site comparisons (ANOVA + Tukey HSD)
# --------------------------------------------------------------

tukey_tidy <- tidy(TukeyHSD(anova)) %>%
  select(comparison = contrast, diff = estimate,
         lower = conf.low, upper = conf.high, p_adj = adj.p.value)

aov_summary <- summary(anova)[[1]]

f_stat <- aov_summary$`F value`[1]
df1    <- aov_summary$Df[1]
df2    <- aov_summary$Df[2]
p_val  <- aov_summary$`Pr(>F)`[1]

table1 <- tukey_tidy %>%
  gt() %>%
  tab_header(
    title = "Table 1. Pairwise comparisons of lc-excess among sampling locations",
    subtitle = "Tukey HSD following one-way ANOVA"
  ) %>%
  fmt_number(columns = c(diff, lower, upper), decimals = 2) %>%
  fmt_number(columns = p_adj, decimals = 4) %>%
  cols_label(
    comparison = "Comparison", diff = "Mean diff.",
    lower = "95% CI (lower)", upper = "95% CI (upper)", p_adj = "Adj. p"
  ) %>%
```

Table 1. Pairwise comparisons of lc-excess among sampling locations

Tukey HSD following one-way ANOVA

Comparison	Mean diff.	95% CI (lower)	95% CI (upper)	Adj. p
it-cc	2.30	0.90	3.69	0.0002
tc-cc	-1.90	-3.06	-0.74	0.0003
utc-cc	-0.75	-1.94	0.43	0.3520
tc-it	-4.19	-5.26	-3.13	0.0000
utc-it	-3.05	-4.14	-1.96	0.0000
utc-tc	1.15	0.37	1.92	0.0010

Overall ANOVA: F(3,111) = 36.53, p < 0.001

```
  gt::tab_source_note(
    source_note = paste0(
      "Overall ANOVA: F(", df1, ",", df2, ") = ", round(f_stat, 2), ", p < 0.001"
    )
  )

table1
```

```
# ------------------------------------------------------------
# TABLE 2. Mixed model: lc-excess ~ location + date (site random effect)
# ------------------------------------------------------------

fixed_loc <- tidy(model_location, effects = "fixed") %>%
  select(term, estimate, std.error, df, statistic, p.value)

rand_loc <- tidy(model_location, effects = "ran_pars") %>%
  select(group, term, estimate)

table2 <- fixed_loc %>%
  gt() %>%
  tab_header(
    title = "Table 2. Linear mixed-effects model of lc-excess",
    subtitle = "Fixed effects: sampling location and date; random intercept: site"
  ) %>%
  fmt_number(columns = c(estimate, std.error, df, statistic), decimals = 3) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
```

Table 2. Linear mixed-effects model of lc-excess

Fixed effects: sampling location and date; random intercept: site

Term	Estimate	SE	df	t	p
(Intercept)	0.286	0.389	25.191	0.735	0.4690
locationit	2.917	0.537	19.550	5.429	0.0000
locationtc	-2.887	0.521	16.754	-5.542	0.0000
locationutc	-1.552	0.549	15.908	-2.828	0.0122
date_c	-0.047	0.005	109.921	-10.179	0.0000

Random effect variance (site) = 0.651; N = 115 observations, 20 sites

```
  cols_label(
    term = "Term", estimate = "Estimate", std.error = "SE",
    df = "df", statistic = "t", p.value = "p"
  ) %>%
  tab_source_note(
    source_note = paste0(
      "Random effect variance (site) = ",
      round(rand_loc$estimate[rand_loc$group == "site"], 3),
      "; N = ", nrow(compiled_isotope), " observations, ",
      n_distinct(compiled_isotope$site), " sites"
    )
  )
table2
```

```
# -------------------------------------------------------------
# TABLE 3. Spatial gradient regressions (lc-excess vs. distance /
# cumulative dams / cumulative dam length), all creeks & dates
# -------------------------------------------------------------

gradient_models <- list(
  list(model = model.1,  creek = "Trail Creek",   date = "6/24", predictor = "Stream distanc
  list(model = model.2,  creek = "Trail Creek",   date = "6/30", predictor = "Stream distanc
  list(model = model.3,  creek = "Trail Creek",   date = "7/15", predictor = "Stream distanc
  list(model = model.4,  creek = "Coal Creek",    date = "6/23", predictor = "Stream distanc
  list(model = model.5,  creek = "Coal Creek",    date = "7/7",  predictor = "Stream distanc
  list(model = model.6,  creek = "Italian Creek", date = "7/1",  predictor = "Stream distanc
  list(model = model.7,  creek = "Italian Creek", date = "7/16", predictor = "Stream distanc
  list(model = model.8,  creek = "Italian Creek", date = "7/23", predictor = "Stream distanc
```

```
  list(model = model.9,  creek = "Trail Creek",   date = "6/24", predictor = "Cumulative dams
  list(model = model.10, creek = "Trail Creek",   date = "6/30", predictor = "Cumulative dams
  list(model = model.11, creek = "Trail Creek",   date = "7/15", predictor = "Cumulative dams
  list(model = model.12, creek = "Coal Creek",    date = "6/23", predictor = "Cumulative dams
  list(model = model.13, creek = "Coal Creek",    date = "7/7",  predictor = "Cumulative dams
  list(model = model.14, creek = "Trail Creek",   date = "6/24", predictor = "Cumulative dam
  list(model = model.15, creek = "Trail Creek",   date = "6/30", predictor = "Cumulative dam
  list(model = model.16, creek = "Trail Creek",   date = "7/15", predictor = "Cumulative dam
  list(model = model.17, creek = "Coal Creek",    date = "6/23", predictor = "Cumulative dam
  list(model = model.18, creek = "Coal Creek",    date = "7/7",  predictor = "Cumulative dam
)

extract_lm <- function(entry) {
  m  <- entry$model
  s  <- summary(m)
  co <- coef(s)
  tibble(
    Predictor   = entry$predictor,
    Creek       = entry$creek,
    Date        = entry$date,
    Intercept   = co[1, 1],
    Slope       = co[2, 1],
    SE          = co[2, 2],
    R2          = s$r.squared,
    p_value     = co[2, 4]
  )
}

table3_df <- map_dfr(gradient_models, extract_lm)

table3 <- table3_df %>%
  gt(groupname_col = "Predictor", rowname_col = "Creek") %>%
  tab_header(
    title = "Table 3. Spatial gradient regressions of lc-excess",
    subtitle = "lc-excess regressed against normalized stream distance, cumulative beaver dar
  ) %>%
  fmt_number(columns = c(Intercept, Slope, SE, R2), decimals = 3) %>%
  fmt_number(columns = p_value, decimals = 4) %>%
  cols_label(
    Date = "Date", Intercept = "Intercept", Slope = "Slope",
    SE = "SE (slope)", R2 = "R\u00b2", p_value = "p"
  ) %>%
```

```r
  tab_options(row_group.as_column = FALSE)

table3
```

```r
# ---------------------------------------------------------------
# TABLE 4. Discharge - lc-excess relationship, upstream vs downstream
# ---------------------------------------------------------------

corr_us <- tidy(cor.test(discharge_utc_isco$discharge, discharge_utc_isco$lc_excess)) %>%
  mutate(Reach = "Upstream")
corr_ds <- tidy(cor.test(discharge_tc_isco$discharge, discharge_tc_isco$lc_excess)) %>%
  mutate(Reach = "Downstream")

lm_us <- tidy(lm(lc_excess ~ discharge, data = discharge_utc_isco)) %>%
  filter(term == "discharge") %>% mutate(Reach = "Upstream")
lm_ds <- tidy(lm(lc_excess ~ discharge, data = discharge_tc_isco)) %>%
  filter(term == "discharge") %>% mutate(Reach = "Downstream")

table4_df <- bind_rows(corr_us, corr_ds) %>%
  select(Reach, r = estimate, conf.low, conf.high, corr_p = p.value, n = parameter) %>%
  mutate(n = n + 2) %>%   # cor.test df = n - 2
  left_join(
    bind_rows(lm_us, lm_ds) %>% select(Reach, slope = estimate),
    by = "Reach"
  )

table4 <- table4_df %>%
  gt() %>%
  tab_header(
    title = "Table 4. Relationship between discharge and lc-excess",
    subtitle = "Trail Creek ISCO stations, Pearson correlation and linear regression slope"
  ) %>%
  fmt_number(columns = c(r, conf.low, conf.high, slope), decimals = 3) %>%
  fmt_number(columns = corr_p, decimals = 4) %>%
  cols_label(
    Reach = "Reach", r = "Pearson r", conf.low = "95% CI (lower)",
    conf.high = "95% CI (upper)", corr_p = "p", n = "n", slope = "Regression slope"
  )
table4
```

Table 3. Spatial gradient regressions of lc-excess

lc-excess regressed against normalized stream distance, cumulative beaver dam count, and cumulative dam length, by creek and sampling date

	Date	Intercept	Slope	SE (slope)	R^2	p
Stream distance (m)						
Trail Creek	6/24	-1.391	-0.001	0.000	0.917	0.0007
Trail Creek	6/30	-2.280	-0.001	0.000	0.893	0.0013
Trail Creek	7/15	-1.103	-0.001	0.000	0.892	0.0014
Coal Creek	6/23	1.539	-0.001	0.000	0.740	0.0614
Coal Creek	7/7	-0.333	-0.001	0.000	0.652	0.0986
Italian Creek	7/1	3.549	0.000	0.000	0.278	0.3617
Italian Creek	7/16	1.438	0.000	0.000	0.837	0.0295
Italian Creek	7/23	1.005	0.000	0.000	0.804	0.0391
Cumulative dams (n)						
Trail Creek	6/24	-1.613	-0.035	0.007	0.867	0.0069
Trail Creek	6/30	-2.501	-0.025	0.005	0.874	0.0062
Trail Creek	7/15	-1.241	-0.033	0.005	0.906	0.0035
Coal Creek	6/23	1.694	-0.047	0.012	0.885	0.0593
Coal Creek	7/7	-2.138	-0.017	0.041	0.074	0.7276
Cumulative dam length (m)						
Trail Creek	6/24	-1.716	-0.056	0.008	0.923	0.0023
Trail Creek	6/30	-2.602	-0.040	0.007	0.884	0.0053
Trail Creek	7/15	-1.349	-0.052	0.006	0.947	0.0011
Coal Creek	6/23	1.548	-0.028	0.004	0.958	0.0214
Coal Creek	7/7	-2.113	-0.011	0.023	0.108	0.6709

Table 4. Relationship between discharge and lc-excess

Trail Creek ISCO stations, Pearson correlation and linear regression slope

Reach	Pearson r	95% CI (lower)	95% CI (upper)	p	n	Regression slope
Upstream	0.609	0.368	0.774	0.0000	40	19.597
Downstream	0.965	0.938	0.981	0.0000	46	42.209

```r
# ---------------------------------------------------------------
# TABLE 5. Pre- vs. post-peak flow lc-excess comparison
# ---------------------------------------------------------------

period_summary <- discharge_utc_isco %>%
  group_by(period) %>%
  summarise(
    n = n(), mean = mean(lc_excess, na.rm = TRUE),
    sd = sd(lc_excess, na.rm = TRUE), median = median(lc_excess, na.rm = TRUE),
    .groups = "drop"
  )

test_results <- bind_rows(
  tidy(t.test(lc_excess ~ period, data = discharge_utc_isco)) %>%
    mutate(Test = "Welch two-sample t-test") %>%
    select(Test, statistic, df = parameter, p.value),
  tidy(wilcox.test(lc_excess ~ period, data = discharge_utc_isco)) %>%
    mutate(Test = "Mann-Whitney U", df = NA_real_) %>%
    select(Test, statistic, df, p.value),
  tidy(var.test(lc_excess ~ period, data = discharge_utc_isco)) %>%
    mutate(Test = "F test (equality of variance)") %>%
    select(Test, statistic, df = num.df, p.value)
)
```

Multiple parameters; naming those columns num.df and den.df.

```r
table5a <- period_summary %>%
  gt() %>%
  tab_header(title = "Table 5a. lc-excess summary statistics by flow period",
             subtitle = "Upstream Trail Creek ISCO (peak flow: 5/14/2026)") %>%
  fmt_number(columns = c(mean, sd, median), decimals = 3) %>%
  cols_label(period = "Period", n = "n", mean = "Mean", sd = "SD", median = "Median")

table5b <- test_results %>%
  gt() %>%
  tab_header(title = "Table 5b. Statistical tests, pre- vs. post-peak lc-excess") %>%
  fmt_number(columns = c(statistic, df), decimals = 2) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
  cols_label(Test = "Test", statistic = "Statistic", df = "df", p.value = "p")

table5a
```

Table 5a. lc-excess summary statistics by flow period

Upstream Trail Creek ISCO (peak flow: 5/14/2026)

Period	n	Mean	SD	Median
post-peak	28	-1.998	0.771	-2.017
pre-peak	12	-0.567	0.333	-0.542

Table 5b. Statistical tests, pre- vs. post-peak lc-excess

Test	Statistic	df	p
Welch two-sample t-test	-8.19	37.96	0.0000
Mann-Whitney U	9.00	NA	0.0000
F test (equality of variance)	5.35	27.00	0.0057

```r
table5b
```

```r
# ------------------------------------------------------------
# TABLE 6. Upstream/downstream mixed model (reach x creek)
# ------------------------------------------------------------

fixed_reach <- tidy(model_reach, effects = "fixed") %>%
  select(term, estimate, std.error, df, statistic, p.value)

anova_reach <- tidy(anova(model_reach)) %>%
  select(term, NumDF = NumDF, DenDF = DenDF, statistic, p.value)
```

```
Warning: The column names NumDF and DenDF in ANOVA output were not recognized or
transformed.
```

```r
table6a <- fixed_reach %>%
  gt() %>%
  tab_header(title = "Table 6a. Fixed effects: lc-excess ~ reach * creek",
             subtitle = "Random intercept: sampling date") %>%
  fmt_number(columns = c(estimate, std.error, df, statistic), decimals = 3) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
  cols_label(term = "Term", estimate = "Estimate", std.error = "SE",
             df = "df", statistic = "t", p.value = "p")
```

Table 6a. Fixed effects: lc-excess ~ reach * creek

Random intercept: sampling date

Term	Estimate	SE	df	t	p
(Intercept)	-1.360	0.805	5.392	-1.689	0.1477
reachupstream	1.901	0.313	5.000	6.075	0.0017
creekItalian Creek	2.753	1.039	5.392	2.650	0.0422
creekTrail Creek	-3.364	1.039	5.392	-3.238	0.0206
reachupstream:creekItalian Creek	-1.108	0.404	5.000	-2.742	0.0407
reachupstream:creekTrail Creek	1.022	0.404	5.000	2.530	0.0526

Table 6b. Type III ANOVA, lc-excess ~ reach * creek

Term	Num. df	Den. df	F	p
reach	1.00	5.00	138.08	0.0001
creek	2.00	5.00	15.43	0.0073
reach:creek	2.00	5.00	17.38	0.0056

```
table6b <- anova_reach %>%
  gt() %>%
  tab_header(title = "Table 6b. Type III ANOVA, lc-excess ~ reach * creek") %>%
  fmt_number(columns = c(NumDF, DenDF, statistic), decimals = 2) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
  cols_label(term = "Term", NumDF = "Num. df", DenDF = "Den. df",
             statistic = "F", p.value = "p")

table6a
```

```
table6b
```

```
# --------------------------------------------------------------
# Save all tables (as individual Word-ready files)
# --------------------------------------------------------------

dir.create("tables", showWarnings = FALSE)
gtsave(table1,  "tables/table1_tukey.docx")
gtsave(table2,  "tables/table2_location_model.docx")
gtsave(table3,  "tables/table3_gradient_models.docx")
```

```r
gtsave(table4,  "tables/table4_discharge.docx")
gtsave(table5a, "tables/table5a_period_summary.docx")
gtsave(table5b, "tables/table5b_period_tests.docx")
gtsave(table6a, "tables/table6a_reach_creek_fixed.docx")
gtsave(table6b, "tables/table6b_reach_creek_anova.docx")
```